STANDARD OPERATING PROCEDURE: HYDRAULIC SYSTEM MAINTENANCE

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Section 1.0 — Purpose & Operational Scope

This procedure establishes mandatory preventive maintenance and fluid replenishment standards for high-pressure hydraulic pumps on Line 3 Assembly. Compliance is mandatory for all reliability technicians.

Section 2.0 — Safety Protocols & Pressure Depressurization

Before servicing pump assemblies, technicians must execute Lockout/Tagout (LOTO) Protocol 4. Ensure reservoir pressure is fully vented to 0 bar via the primary manual relief valve.

Section 3.0 — Wear Inspection & Seal Tolerances

High-pressure elastomer seals (Part AFD-SEAL-882) must be inspected for thermal micro-fissures and durometer degradation. Any seal showing radial cracking must be replaced immediately.

Section 4.0 — Fluid Evacuation & Flush Protocols

Hydraulic fluid degradation is the primary driver of premature seal blowout and pump cavitation. Standard mineral oils oxidize rapidly under peak assembly cycles.

Section 4.2 — Mandatory Synthetic Oil Flush Intervals

Crucial Preventive Measure: Pumps operating above 180 bar must undergo complete fluid evacuation and replenishment with high-grade synthetic ester hydraulic fluid every **250 operational hours**. Historical telemetry demonstrates that implementing quarterly synthetic flushes reduced similar pump cavitation and seal breakdown downtime by **41% during 2023**.